FULL-STACK ARCHITECTURE & TECHNICAL OVERVIEW

BiteRush

Next-Gen Real-Time Food Delivery & Restaurant Platform

Author / Developer: Vaibhav Singh **Architecture:** MERN + Socket.IO + Cloud Services

Version: Production 1.0 **Date:** October 2026

1. Executive Overview

BiteRush is a modern, enterprise-grade food delivery web application that bridges three distinct user groups: **Customers**, **Restaurant Owners**, and **Delivery Partners (Riders)** into one unified, real-time ecosystem.

Customer

ORDERING

Location detection, smart restaurant discovery, dynamic cart management, Razorpay checkout, live socket tracking, and OTP verification.

Restaurant Owner

MANAGEMENT

Shop creation, menu & food item management, price controls, Cloudinary image assets, incoming order processing, and dispatch alerts.

Delivery Partner

LOGISTICS

Proximity-based order feeds, GPS broadcast via WebSockets, Leaflet route navigation, OTP collection, and proof of delivery.

2. Complete Technology Stack ("What is used for what")

A transparent, component-by-component breakdown of every technology, library, and cloud service implemented in the BiteRush codebase:

TECHNOLOGY / TOOL	CATEGORY	EXACT PURPOSE & VALUE IN BITERUSH
`React 19`	FRONTEND	Core frontend UI library providing reusable component architecture, fast virtual DOM reconciliation, and fluid single-page user experiences.
`Vite 7`	BUILD TOOL	Next-generation development server and production bundler, offering lightning-fast Hot Module Replacement (HMR) and optimized Rollup builds.
`Tailwind CSS v4`	STYLING	Utility-first CSS framework delivering modern styling, custom warm food palettes (Tomato Red `#E63946`, Warm Coral `#FF6B5E`, Soft Ivory), and responsive layouts.
`Redux Toolkit`	STATE MGMT	Centralized predictable state store managing global application state: user authentication, active role, current GPS coordinates, and real-time cart item synchronization.
`React Router DOM v7`	ROUTING	Client-side declarative routing handling navigation between landing, auth, customer dashboards, checkout, rider portal, shop administration, and order tracking.
`Node.js & Express.js`	BACKEND API	High-performance asynchronous JavaScript runtime and web framework powering REST API routes, controllers, middleware, and request validation.
`MongoDB Atlas`	DATABASE	Cloud-native NoSQL document database providing resilient, schema-flexible storage for user accounts, restaurant profiles, food catalogs, and orders.
`Mongoose`	ODM	Object Data Modeling library providing strict schema validation, type casting, model hooks, document relationships, and geospatial/index optimizations.
`Socket.IO`	REAL-TIME	Full-duplex, bidirectional WebSocket engine enabling live rider GPS location broadcasting, real-time order status transitions, and instant customer tracking updates.

TECHNOLOGY / TOOL	CATEGORY	EXACT PURPOSE & VALUE IN BITERUSH
`JSON Web Token (JWT)`	SECURITY	Stateless, tamper-proof authentication mechanism issuing digitally signed HTTP-only session cookies and authorizing secure Socket.IO connection handshakes.
`Bcrypt.js`	SECURITY	Adaptive cryptographic password-hashing function using salt rounds to guarantee user passwords are never stored in plain text.
`Firebase Auth`	AUTH	Enables frictionless Google OAuth One-Tap sign-in and sign-up, securely bridging Google user tokens with backend database user records.
`Razorpay Gateway`	PAYMENTS	Fintech payment SDK handling end-to-end online transactions (UPI, Credit/Debit cards, Netbanking) with server-side order generation and cryptographic HMAC signature validation.
`Cloudinary`	MEDIA STORAGE	Cloud-based asset storage and CDN pipeline that stores restaurant banners and food dishes, automatically compressing images and generating optimized responsive WebP URLs.
`Brevo (Sendinblue)`	EMAIL SERVICE	High-reputation transactional email API service that delivers automated password reset OTPs and delivery completion verification OTPs directly to user inboxes.
`Leaflet & React-Leaflet`	INTERACTIVE MAPS	Lightweight, mobile-friendly OpenStreetMap engine rendering dynamic map tiles, live delivery markers, custom scooter icons, and interactive visual routes.
`Geoapify API`	GEOCODING	Reverse-geocoding service translating GPS coordinates (Latitude/Longitude) into clean human-readable street names, localities, and cities.
`Axios`	HTTP CLIENT	Promise-based HTTP client for browser requests, configured with `withCredentials: true` to automatically convey secure HTTP-only cookies across requests.
`Multer`	FILE UPLOAD	Node.js middleware handling `multipart/form-data` requests to temporarily stage restaurant and dish photos on the server before dispatching to Cloudinary.

3. Critical Architectural Workflows

How individual services interconnect to form reliable end-to-end user journeys:

A. Live Order Lifecycle & Real-Time Socket.IO Synchronization

1. **Customer Places Order:** Cart items are validated, Razorpay signature or Cash on Delivery verified, and order saved with status `"order placed"`.
2. **Proximity BroadCast:** Backend emits Socket.IO event to nearby Delivery Partners matching the restaurant locality.
3. **Rider Accepts Assignment:** Rider accepts the delivery task; status flips to `"preparing"` and then `"out for delivery"`. Customer's screen instantly updates without reloading.
4. **Continuous GPS Streaming:** Rider's phone emits `updateDeliveryBoyLocation` at regular intervals; customer's Leaflet map marker moves in real-time along the route.
5. **Two-Factor OTP Handover:** A 4-digit OTP is generated and emailed to the customer via Brevo API. When rider reaches the destination, customer gives OTP; rider submits it, and status seals to `"delivered"`.

B. High-Res Media Pipeline (Cloudinary with Smart Local Fallback)

1. **Upload Submission:** Owner uploads photo from desktop/phone via standard file input.
2. **Multer Staging:** Backend parses `multipart/form-data` into temporary buffer in `public/` folder.
3. **Cloud Upload & CDN Delivery:** Photo is pushed directly to Cloudinary Media Asset bucket with automatic optimization and stored as a lightweight HTTPS CDN URL in MongoDB.
4. **Resilient Fallback:** If cloud keys are missing during local development, an automated Base64 fallback ensures the application never crashes and continues running seamlessly.

4. Enterprise Security & Engineering Practices

Socket Hijack Prevention

SOCKET SECURITY

Socket.IO connection handshake validates verified JWT authentication tokens. Users can only listen to and broadcast updates for orders belonging strictly to their identity.

Idempotent Payments

RAZORPAY

Cryptographic HMAC SHA256 signature verification guarantees transactions cannot be spoofed, preventing double-debiting and duplicate order fulfillment.

Role-Based Access Control

RBAC

Strict authorization middleware restricts administrative actions (e.g., editing shop menus, changing prices, completing deliveries) strictly to authenticated owners or riders.

Production Deployment Ready

VERCEL & RENDER

Pre-configured `vercel.json` rewrites for Single Page Application routing, Git automated CI/CD workflows, and environment variable isolation for production stability.